CEE6501 — Lecture 8.1

Member Releases

Plane frames and beams with moment releases (hinges)

Learning Objectives

By the end of this lecture, you will be able to:

- Explain what a **member release** is and why it matters in DSM
- Classify members by **release type** (no hinge / hinge at b / hinge at e / hinges at both ends)
- Derive modified local stiffness relations by enforcing **zero end moment** at a hinge
- Write the resulting **modified local stiffness matrix** and **modified fixed-end force vector**
- Recognize and handle **hinged joints** and why they can cause singular stiffness matrices

Agenda

- **Part 1** — Concepts + notation
- **Part 2** — Derivations (MT = 1, 2, 3)
- **Part 3** — Hinged joints + singularity
- **Part 4** — Wrap-up

Part 1 — Concepts and Notation

What is a member release?

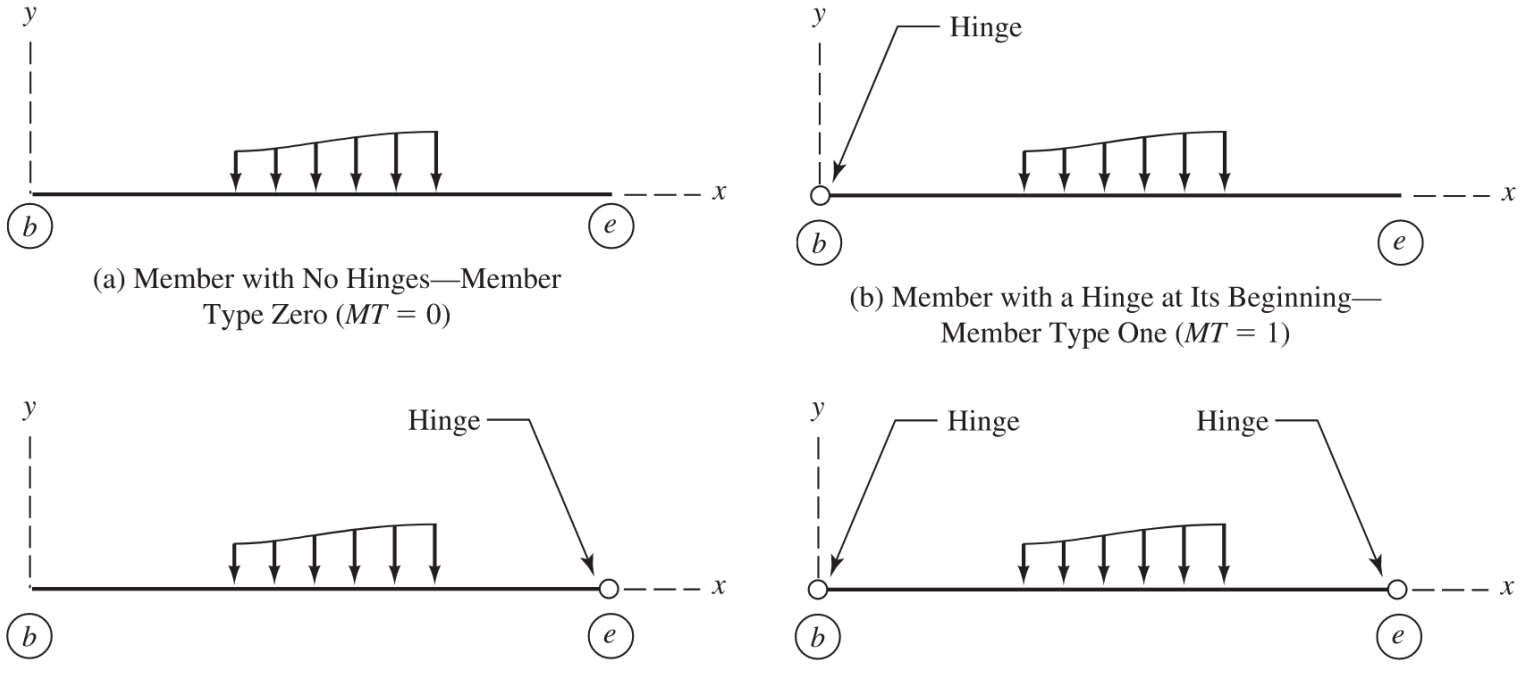
- A **member release** removes a force/moment transfer capability at an end of a member.
- Here we focus on **moment releases** (hinges).
- If an end is hinged, the **end moment must be zero** at that end.

Key constraint (local):

$$M_b = 0 \quad \text{or} \quad M_e = 0$$

Member Release Types (Classification)

- **MT = 0:** no hinges (rigid at both ends, normal frame element)
- **MT = 1:** hinge at **beginning** (end **b**)
- **MT = 2:** hinge at **end** (end **e**)
- **MT = 3:** hinges at **both ends**



(a) Member with No Hinges—Member Type Zero ($MT = 0$)

(b) Member with a Hinge at Its Beginning—Member Type One ($MT = 1$)

(c) Member with a Hinge at Its End—Member Type Two ($MT = 2$)

(d) Member with Hinges at Both Ends—Member Type Three ($MT = 3$)

Reminder: Frame Local DOFs and End Forces Notation

Local displacement vector (member coordinates):

$$\mathbf{u}^T = \begin{bmatrix} u_1 & u_2 & u_3 & u_4 & u_5 & u_6 \end{bmatrix} = \begin{bmatrix} u_b & v_b & \theta_b & u_e & v_e & \theta_e \end{bmatrix}$$

Local end-force vector:

$$\mathbf{Q}^T = \begin{bmatrix} Q_1 & Q_2 & Q_3 & Q_4 & Q_5 & Q_6 \end{bmatrix} = \begin{bmatrix} N_b & V_b & M_b & N_e & V_e & M_e \end{bmatrix}$$

Reminder: Frame Fixed-End Force Notation (FEFs)

We write local fixed-end forces using the convention:

- Q^F_{xb} = FEF at node **b** in local **x** (axial) → corresponds to **DOF 1**
- Q^F_{yb} = FEF at node **b** in local **y** (shear) → corresponds to **DOF 2**
- Q^F_{mb} = FEF at node **b** (moment) → corresponds to **DOF 3**
- Q^F_{xe} = FEF at node **e** in local **x** (axial) → corresponds to **DOF 4**
- Q^F_{ye} = FEF at node **e** in local **y** (shear) → corresponds to **DOF 5**
- Q^F_{me} = FEF at node **e** (moment) → corresponds to **DOF 6**

So the fixed-end force vector is

$$\mathbf{Q}^{F,T} = \begin{bmatrix} Q^F_{xb} & Q^F_{yb} & Q^F_{mb} & Q^F_{xe} & Q^F_{ye} & Q^F_{me} \end{bmatrix}^T$$

Part 2 — Derivations for Released Members

Starting point (MT = 0): standard local stiffness relation

$$\mathbf{Q} = \mathbf{k}\,\mathbf{u} + \mathbf{Q}^F$$

Member releases modify **only**:

- the local stiffness matrix $\mathbf{k}$, and
- the fixed-end force vector $\mathbf{Q}^F$

so the DSM workflow stays the same.

Base local stiffness relations (no hinges)

Recall from last lecture. For a plane frame member with no hinges, the local end-force components can be written (explicitly) as 6 independent equations:

$$
\begin{aligned}
Q_1 &= \frac{EA}{L}(u_1 - u_4) + Q_{xb}^F \\
Q_2 &= \frac{EI}{L^3}(12u_2 + 6Lu_3 - 12u_5 + 6Lu_6) + Q_{yb}^F \\
Q_3 &= \frac{EI}{L^3}(6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6) + Q_{mb}^F \\
Q_4 &= \frac{EA}{L}(-u_1 + u_4) + Q_{xe}^F \\
Q_5 &= \frac{EI}{L^3}(-12u_2 - 6Lu_3 + 12u_5 - 6Lu_6) + Q_{ye}^F \\
Q_6 &= \frac{EI}{L^3}(6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6) + Q_{me}^F
\end{aligned}
$$

6 Independent Equations in Matrix Form

$$\begin{Bmatrix} Q_1 \\ Q_2 \\ Q_3 \\ Q_4 \\ Q_5 \\ Q_6 \end{Bmatrix} = \begin{bmatrix} k_{11} & k_{12} & k_{13} & k_{14} & k_{15} & k_{16} \\ k_{21} & k_{22} & k_{23} & k_{24} & k_{25} & k_{26} \\ k_{31} & k_{32} & k_{33} & k_{34} & k_{35} & k_{36} \\ k_{41} & k_{42} & k_{43} & k_{44} & k_{45} & k_{46} \\ k_{51} & k_{52} & k_{53} & k_{54} & k_{55} & k_{56} \\ k_{61} & k_{62} & k_{63} & k_{64} & k_{65} & k_{66} \end{bmatrix} \begin{Bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \\ u_5 \\ u_6 \end{Bmatrix} + \begin{Bmatrix} Q_1^F \\ Q_2^F \\ Q_3^F \\ Q_4^F \\ Q_5^F \\ Q_6^F \end{Bmatrix}$$

MT = 0: Matrix Form

$$\mathbf{k}_{\text{frame}} = \begin{bmatrix} \frac{EA}{L} & 0 & 0 & -\frac{EA}{L} & 0 & 0 \\ 0 & \frac{12EI}{L^3} & \frac{6EI}{L^2} & 0 & -\frac{12EI}{L^3} & \frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{4EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{2EI}{L} \\ -\frac{EA}{L} & 0 & 0 & \frac{EA}{L} & 0 & 0 \\ 0 & -\frac{12EI}{L^3} & -\frac{6EI}{L^2} & 0 & \frac{12EI}{L^3} & -\frac{6EI}{L^2} \\ 0 & \frac{6EI}{L^2} & \frac{2EI}{L} & 0 & -\frac{6EI}{L^2} & \frac{4EI}{L} \end{bmatrix}$$

MT = 1 (hinge at b): enforce $M_b = 0$

Hinge at the beginning means

$$M_b = Q_3 = 0$$

Start from

$$Q_3 = \frac{EI}{L^3}(6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6) + Q_{mb}^F$$

Set $Q_3 = 0$ and solve for u_3 (released rotation at b).

MT = 1: solve for the released rotation u_3

Set $Q_3 = 0$:

$$0 = \frac{EI}{L^3}(6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6) + Q_{mb}^F$$

Multiply by $\frac{L^3}{EI}$ and isolate u_3:

$$4L^2u_3 = -6L(u_2 - u_5) - 2L^2u_6 - \frac{L^3}{EI}Q_{mb}^F$$

$$u_3 = -\frac{3}{2L}(u_2 - u_5) - \frac{1}{2}u_6 - \frac{L}{4EI}Q_{mb}^F$$

Key point: *u_3 is no longer an independent DOF in the member relations. The row/column corresponding to this DOF can be expressed in terms of a linear combination of other rows/columns in the matrix.*

MT = 1: substitute u_3 back into Q_2, Q_5, Q_6

Resulting modified equations (MT = 1):

$$
\begin{aligned}
Q_1 &= \frac{EA}{L}(u_1 - u_4) + Q_{xb}^F \\
Q_2 &= \frac{EI}{L^3}(3u_2 - 3u_5 + 3Lu_6) + \left(Q_{yb}^F - \frac{3}{2L}Q_{mb}^F\right) \\
Q_3 &= 0 \\
Q_4 &= \frac{EA}{L}(-u_1 + u_4) + Q_{xe}^F \\
Q_5 &= \frac{EI}{L^3}(-3u_2 + 3u_5 - 3Lu_6) + \left(Q_{ye}^F + \frac{3}{2L}Q_{mb}^F\right) \\
Q_6 &= \frac{EI}{L^3}\left(3Lu_2 - 3Lu_5 + 3L^2u_6\right) + \left(Q_{me}^F - \frac{1}{2}Q_{mb}^F\right)
\end{aligned}
$$

MT = 1: stiffness matrix

row/column 3 are 0's:

$$\mathbf{k}_{(MT=1)} = \frac{EI}{L^3}\begin{bmatrix} \frac{AL^2}{I} & 0 & \boxed{0} & -\frac{AL^2}{I} & 0 & 0 \\ 0 & 3 & \boxed{0} & 0 & -3 & 3L \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \\ -\frac{AL^2}{I} & 0 & \boxed{0} & \frac{AL^2}{I} & 0 & 0 \\ 0 & -3 & \boxed{0} & 0 & 3 & -3L \\ 0 & 3L & \boxed{0} & 0 & -3L & 3L^2 \end{bmatrix}$$

MT = 1: fixed end force vector

row/column 3 are 0's:

$$\mathbf{Q}^F_{(MT=1)} = \begin{bmatrix} F^F_{xb} \\ F^F_{yb} - \frac{3}{2L} F^F_{mb} \\ \boxed{0} \\ F^F_{xe} \\ F^F_{ye} + \frac{3}{2L} F^F_{mb} \\ F^F_{me} - \frac{1}{2} F^F_{mb} \end{bmatrix}$$

MT = 2 (hinge at e): enforce $M_e = 0$

Hinge at the end means

$$M_e = Q_6 = 0$$

Start from

$$Q_6 = \frac{EI}{L^3}(6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6) + Q_{me}^F$$

Set $Q_6 = 0$ and solve for u_6 (released rotation at e).

MT = 2: solve for the released rotation u_6

Set $Q_6 = 0$:

$$0 = \frac{EI}{L^3}(6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6) + Q_{me}^F$$

Multiply by $\frac{L^3}{EI}$ and isolate u_6:

$$4L^2u_6 = -6L(u_2 - u_5) - 2L^2u_3 - \frac{L^3}{EI}Q_{me}^F$$

$$u_6 = -\frac{3}{2L}(u_2 - u_5) - \frac{1}{2}u_3 - \frac{L}{4EI}Q_{me}^F$$

MT = 2: substitute u_6 back into Q_2, Q_3, Q_5

Resulting modified equations (MT = 2):

$$
\begin{aligned}
Q_1 &= \frac{EA}{L}(u_1 - u_4) + Q_{xb}^F \\
Q_2 &= \frac{EI}{L^3}(3u_2 + 3Lu_3 - 3u_5) + \left(Q_{yb}^F - \frac{3}{2L}Q_{me}^F\right) \\
Q_3 &= \frac{EI}{L^3}\left(3Lu_2 + 3L^2u_3 - 3Lu_5\right) + \left(Q_{mb}^F - \frac{1}{2}Q_{me}^F\right) \\
Q_4 &= \frac{EA}{L}(-u_1 + u_4) + Q_{xe}^F \\
Q_5 &= \frac{EI}{L^3}(-3u_2 - 3Lu_3 + 3u_5) + \left(Q_{ye}^F + \frac{3}{2L}Q_{me}^F\right) \\
Q_6 &= 0
\end{aligned}
$$

MT = 2: stiffness matrix

Row and column 6 are zero:

$$\mathbf{k}_{(MT=2)} = \frac{EI}{L^3}\begin{bmatrix} \frac{AL^2}{I} & 0 & 0 & -\frac{AL^2}{I} & 0 & \boxed{0} \\ 0 & 3 & 3L & 0 & -3 & \boxed{0} \\ 0 & 3L & 3L^2 & 0 & -3L & \boxed{0} \\ -\frac{AL^2}{I} & 0 & 0 & \frac{AL^2}{I} & 0 & \boxed{0} \\ 0 & -3 & -3L & 0 & 3 & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \end{bmatrix}$$

MT = 2: fixed-end force vector

Row/column 6 are zero:

$$\mathbf{Q}_{f,(MT=2)} = \begin{bmatrix} Q_{xb}^F \\ Q_{yb}^F - \dfrac{3}{2L} Q_{me}^F \\ Q_{mb}^F - \dfrac{1}{2} Q_{me}^F \\ Q_{xe}^F \\ Q_{ye}^F + \dfrac{3}{2L} Q_{me}^F \\ \boxed{0} \end{bmatrix}$$

MT = 3 (hinges at both ends): enforce $M_b = 0$ and $M_e = 0$

Hinges at both ends mean

$$Q_3 = 0, \qquad Q_6 = 0$$

We now have **two constraints** that couple u_3 and u_6.

Start from the base moment equations:

$$Q_3 = \frac{EI}{L^3}(6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6) + Q_{mb}^F$$

$$Q_6 = \frac{EI}{L^3}(6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6) + Q_{me}^F$$

MT = 3: solve the coupled system for u_3 and u_6

Set $Q_3 = 0$ and $Q_6 = 0$, multiply both by $\frac{L^3}{EI}$:

$$0 = 6Lu_2 + 4L^2u_3 - 6Lu_5 + 2L^2u_6 + \frac{L^3}{EI}Q^F_{mb}$$

$$0 = 6Lu_2 + 2L^2u_3 - 6Lu_5 + 4L^2u_6 + \frac{L^3}{EI}Q^F_{me}$$

Solving gives (one convenient form):

$$u_3 = -\frac{1}{L}(u_2 - u_5) - \frac{L}{6EI}(2Q^F_{mb} - Q^F_{me})$$

$$u_6 = -\frac{1}{L}(u_2 - u_5) - \frac{L}{6EI}(2Q^F_{me} - Q^F_{mb})$$

MT = 3: substitute u_3 and u_6 back Q_2, Q_5

$$
\begin{aligned}
Q_1 &= \frac{EA}{L}(u_1 - u_4) + Q_{xb}^F \\
Q_2 &= Q_{yb}^F - \frac{1}{L}(Q_{mb}^F + Q_{me}^F) \\
Q_3 &= 0 \\
Q_4 &= \frac{EA}{L}(-u_1 + u_4) + Q_{xe}^F \\
Q_5 &= Q_{ye}^F + \frac{1}{L}(Q_{mb}^F + Q_{me}^F) \\
Q_6 &= 0
\end{aligned}
$$

Interpretation

- Notice that Q_2 **and** Q_5 **no longer contain any displacement terms**.
- They are **purely functions of the fixed-end forces**.
- The bending stiffness contributions have completely disappeared.
- This means the element **cannot develop bending resistance**.
- The transverse shear forces are determined solely by equilibrium of the fixed-end moments.
- The element now behaves as a **pure axial (truss-type) member**.

MT = 3: stiffness matrix

Not only are row/column 3 and 6 are zero, but also row/column 2 and 5:

$$\mathbf{k}_{(MT=3)} = \frac{EA}{L}\begin{bmatrix} 1 & 0 & \boxed{0} & -1 & 0 & \boxed{0} \\ 0 & 0 & \boxed{0} & 0 & 0 & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \\ -1 & 0 & \boxed{0} & 1 & 0 & \boxed{0} \\ 0 & 0 & \boxed{0} & 0 & 0 & \boxed{0} \\ \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} & \boxed{0} \end{bmatrix}$$

What does this stiffness matrix remind you of?

MT = 3: fixed-end force vector

Row/column 3 and 6 are zero:

$$\mathbf{Q}^F_{(MT=3)} = \begin{bmatrix} Q^F_{xb} \\ Q^F_{yb} - \frac{1}{L}(Q^F_{mb} + Q^F_{me}) \\ \boxed{0} \\ Q^F_{xe} \\ Q^F_{ye} + \frac{1}{L}(F^F_{mb} + Q^F_{me}) \\ \boxed{0} \end{bmatrix}$$

Implementation note: keep 6×6, but released DOFs have zero rows/cols

- Even though releases reduce the **independent** member coordinates, we often keep a **6×6** member form.
- Released moment rows/cols (and their corresponding fixed-end terms) become **zero** in the stiffness matrix and FEF vector.
- This keeps the DSM assembly code clean: same DOF mapping size for every member.

Then transform each element to global coordinates as usual:

$$\mathbf{K} = \mathbf{T}^T \mathbf{k} \mathbf{T}, \qquad \mathbf{F}^F = \mathbf{T}^T \mathbf{Q}^F$$

Part 3 — Hinged Joints and Matrix Singularity

Hinged joints in beams and plane frames

A **hinged joint** is a joint where all members meeting at that joint are connected by **moment releases**.

Figure placeholder: Fig. 7.2 — Example plane frame showing a hinged joint

Physical consequence:

- The joint **cannot transmit bending moment**.
- The joint is **free to rotate**.
- Axial and shear forces may still be transmitted.

What does this mean for the DOFs?

At a hinged joint:

- The rotational DOF θ_j is not constrained (it is free).
- But no connected member provides **rotational stiffness**.
- The joint rotation does not generate bending moment.

In other words: The structure has a rotation that costs **zero strain energy**.

DSM partitioned equilibrium

Recall the partitioned system:

$$\begin{bmatrix} \mathbf{K}_{ff} & \mathbf{K}_{fr} \\ \mathbf{K}_{rf} & \mathbf{K}_{rr} \end{bmatrix} \begin{bmatrix} \mathbf{u}_f \\ \mathbf{u}_r \end{bmatrix} = \begin{bmatrix} \mathbf{F}_f \\ \mathbf{F}_r \end{bmatrix}$$

We solve for the unknown displacements from

$$\mathbf{u}_f = K_{ff}^{-1}(\mathbf{F}_f - \mathbf{K}_{fr}\mathbf{u}_r)$$

This requires that $\mathbf{K}_{ff}$ be **invertible**.

What happens if θ_j is left in the free set?

The rotational DOF θ_j is not restrained, so it is placed in the free displacement vector $\mathbf{u}_f$.

As a free DOF, it has a corresponding row and column in the stiffness matrix $\mathbf{K}_{ff}$.

However, at a fully hinged joint, no connected element provides rotational stiffness. From our earlier derivations, the entire row and column associated with θ_j are zero.

Therefore:

- $\mathbf{K}_{ff}$ contains a **zero row and column**.
- $\mathbf{K}_{ff}$ becomes **singular** (non-invertible).
- The system of equations does not admit a unique displacement solution.

Important clarification

This situation does **not** indicate a physical instability.

The structure may be completely stable in translation and capable of carrying load.

The singularity arises because a rotational DOF was left in the free set **without any associated stiffness**.

In other words:

- The joint can rotate freely.
- That rotation does not generate internal moment.
- The stiffness matrix has no resistance in that direction.

The issue is therefore **mathematical**, not structural.

To obtain a well-posed system of equations, this DOF must be treated properly in the DSM formulation.

Two Modeling Approaches

Two practical ways to avoid singularity:

Method A — Artificially restrain (fix) the joint rotation

- Put the hinged-joint rotation into the **restrained set** $\mathbf{u}_r$ (not in $\mathbf{u}_f$)
- The computed reaction moment at that DOF ends up being **zero** anyway

Method B — Artifically stiffen the joint DOF

- Model the joint as rigidly connected to **one** of the members meeting there
- Assign the joint three DOFs (u_x, u_y, θ) as usual
- Other members connect by hinges (moment releases) so the net joint moment remains zero

Part 4 - Method A, Fictitious Fixity

The Main Idea

The most straightforward remedy for the singularity is:

> *Eliminate the rotational DOF of hinged joints by modeling such joints as restrained (fixed) against rotation.*

This is imposing **fictitious fixity** that does not change global structural behavior.

Internal Moment at a Fully Hinged Joint

At a fully hinged joint:

- All connected members have **moment releases** at the node.
- The joint **cannot transmit bending moment**.
- Therefore, each connected element, i, contributes

$$M_i = 0 \quad \text{at the joint.}$$

Joint equilibrium must always satisfy

$$\sum_{i=1}^{n_{ele}} M_i + M_{\text{ext}} = 0$$

where:

- M_i = internal end moment from element i
- M_{ext} = applied or reaction moment at the joint

Artificially Fixing the Rotation

Suppose we introduce **fictitious fixity** by restraining the rotation at the joint DOF associated with member j:

$$\theta_j = 0$$

Instead of leaving θ_j in the free displacement vector, we move it into the restrained set.

After solving for the free DOFs, reactions F_r are computed at all restrained DOFs. For the fictitiously fixed rotation for member j, the reported reaction is a moment:

$$M_{\text{ext}} = M_j$$

What Happens to the "Reaction" at That DOF?

As stated, joint equilibrium requires

$$\sum_{i=1}^{n_{ele}} M_i + M_{\text{ext}} = 0$$

Separating the contribution from member j from the internal moment sum and setting $M_{\text{ext}} = M_j$.

$$\left(\sum_{\substack{i=1 \\ i \neq j}}^{n_{ele}} M_i \right) + M_j = 0.$$

At a hinged joint, all other connected members contribute zero moment, so $M_j = 0$

Thus, the reported "reaction" at the fictitiously fixed member, j, is a numerical artifact and evaluates to zero (up to roundoff).

Why the Structural Response is Unchanged

Imposing $\theta_j = 0$ does not change the moments developed in the joint, because:

- No connected element provides rotational stiffness.
- The global stiffness matrix contains no coupling terms involving θ_j.
- The equilibrium equations do not depend on that rotational DOF.

Because no bending stiffness exists at the joint, fixing the rotation does not generate any additional reaction moment (moment is related to displacement through stiffness).

You can think of DOF j as a restrained DOF that simply develops **no reaction load**.

Implementing with DSM Partitioning

Instead of leaving θ_j in the free displacement vector $\mathbf{u}_f$,

we classify it as a **restrained DOF**.

So:

- It is placed in $\mathbf{u}_r$.
- It is removed from $\mathbf{K}_{ff}$.
- It is no longer part of the system we invert:

$$\mathbf{K}_{ff}^{-1}$$

The zero-stiffness row and column disappear from the matrix being solved.

Method A Summary

Method A is therefore a **numerical bookkeeping device**:

- It removes a zero-stiffness DOF from the unknown set,
- Eliminates the singularity in $\mathbf{K}_{ff}$,
- Preserves the correct internal force distribution.

The physics is unchanged but the algebra becomes well-posed.

Method A — Step-by-Step Procedure

1. **Identify the hinged joint**

   - All connected member ends at that node are moment-released.
   - The joint cannot transmit bending moment.

2. **Classify the rotation as a restrained DOF**

   - Place θ_j in the restrained displacement vector $\mathbf{u}_r$.
   - Impose the artificial condition

$$\theta_j = 0$$

This removes the hinge rotation from the unknown set, u_f.

3. **Assemble the global stiffness matrix normally**

    - Because all connected elements are released, the assembled stiffness terms associated with θ_j are zero.
    - No rotational stiffness is introduced.

4. **Solve for the free DOFs**

    - θ_j is not included in $\mathbf{u}_f$.
    - The matrix $\mathbf{K}_{ff}$ no longer contains a zero row/column.
    - The system becomes invertible.

5. **Compute reactions**

    Reactions are obtained from

$$\mathbf{F}_r = \mathbf{K}_{rf}\mathbf{u}_f + \mathbf{K}_{rr}\mathbf{u}_r + \mathbf{F}_r^F$$

    The reaction moment at the fictitiously fixed DOF evaluates to

$$M_j \approx 0$$

    confirming hinge behavior.

Part 5 - Method B, Fictitious Stiffness

The Main Idea

Instead of removing the joint rotation from the free system (Method A), we:

> *Model **one (and only one)** of the members meeting at that joint as having rotational rigidity.*

All other members framing into the joint remain **hinged (moment-released)**.

Figure placeholder: Joint j rigid to one member, hinged to others

Why This Removes the Singularity

By making **one member rigidly connected**, that member contributes:

$$\frac{4EI}{L}$$

type rotational stiffness at the joint.

Therefore:

- The joint rotation DOF now has stiffness.
- $\mathbf{K}_{ff}$ is no longer singular.
- The system becomes solvable without fictitious restraints.

Why This Does NOT Change the Structural Behavior

Even though one member, j, is fictitiously modeled as rigid at the joint:

- There is **no external moment applied at that joint**.
- All other members connected there are hinged.
- Hinged ends cannot transmit moment.

From joint equilibrium:

$$\sum_{i=1}^{n_{ele}} M_i = 0$$

All hinged members satisfy $M_i = 0$, so the joint moment equilibrium can be written as:

$$\sum_{i=1}^{n_c} M_i = \left(\sum_{\substack{i=1 \\ i \neq j}}^{n_c} M_i \right) + M_j = 0 + M_j = 0.$$

Thus, the rigidly connected member must also have zero end moment, $M_j = 0$, in the final solution in order to satisfy joint equilibrium.

Key Mechanical Insight

Although we introduced rotational stiffness at the joint:

- The structure has no moment demand at that location.
- Static equilibrium forces the joint moment to zero.
- The internal force distribution remains unchanged.

We have:

- Stabilized the algebra (no singular matrix),
- Preserved correct statics,
- Avoided artificial restraints.

What Method B Really Is

Method B is a **modeling device** that:

- Supplies stiffness to a mechanism DOF,
- Keeps the joint rotation as a free unknown,
- Relies on equilibrium to enforce zero moment at the joint (i.e., correct hinge behavior).

The structural response remains the same. The numerical system becomes well-posed.

Method B — Step-by-Step Procedure

1. **Identify the fully hinged joint**
    - Confirm that all members meeting at the joint are moment-released.
    - Recognize that this creates a rotational DOF with zero stiffness.
2. **Select one "reference" member at the joint**
    - Choose **one (and only one)** connected member.
    - Model that member as **rigidly connected** at the joint (no moment release at that end).
    - The rotation DOF is now supported by this member's rotational stiffness.

    This supplies stiffness to the otherwise unsupported rotation.
3. **Keep all other members hinged at that joint**
    - Use the appropriate released-end formulation (MT = 1 or MT = 2).
    - Their end moments at the joint are explicitly zero.

    These members do **not** contribute rotational stiffness at the joint.

4. **Treat the joint as a normal rigid node in global bookkeeping**

    - The rotational DOF θ_j remains in the **free displacement vector**.
    - Because one member now contributes stiffness,
      the corresponding row/column in $\mathbf{K}_{ff}$ is no longer zero.

    The singularity disappears.

5. **Assemble and solve the system**

    - Assemble the global stiffness matrix in the usual way.
    - Solve for the free DOFs.
    - Compute member end forces.

    Although one member was modeled as rigid at the joint,
    equilibrium at that joint still requires:

$$\sum M = 0$$

    Since the other members are hinged and no external moment is applied,
    the rigid member's end moment evaluates to:

$$M_j \approx 0$$

Method A vs Method B — Clear Takeaway

Method A — Fictitious Fixity

- Reclassify the joint rotation DOF as **restrained**.

Effect:

- Eliminates the zero-stiffness DOF from $\mathbf{K}_{ff}$.
- The computed fictituous reaction moment evaluates to $M_j = 0$

Method B — Fictitious Stiffness

- Keep the joint rotation DOF **free**.
- Make **one (and only one)** connected member rigid at the joint.
- All other connected members remain hinged.

Effect:

- The rigid member supplies nonzero rotational stiffness.
- From joint equilibrium, $\sum M = 0$, the fictituously rigid member's end moment evaluates to zero.

What Is Different?

- **Method A:** enforces $\theta_j = 0$
- **Method B:** allows $\theta_j \neq 0$ (determined numerically from solving u_f)

Therefore, local element rotations can be slightly different across methods, but this does not affect the internal force distribution.

Key Takeaway

Both methods represent the **same physical structure** and satisfy the same equilibrium and compatibility conditions.

- Internal force distribution → **the same**
- Physically meaningful deformations → **the same**
- Reported joint rotation at the hinges → **may differ**

The difference is numerical, not structural.

Part 5 — Implementing Method A

Problem data (from the example)

Assume member properties are constant for all members:

- $E = 200 \times 10^6 \text{ kN/m}^2 \ (= 200 \text{ GPa})$
- $A = 0.0065 \text{ m}^2 \ (= 6500 \text{ mm}^2)$
- $I = 150 \times 10^{-6} \text{ m}^4 (= 150 \times 10^6 \text{ mm}^4)$
- $L = 5 \text{ m}$

Local DOF order for each 2D frame member:

$$[u_{xb},\ u_{yb},\ \theta_b,\ u_{xe},\ u_{xe},\ \theta_e]$$

Fixed-end force notation:

$$\begin{aligned} \text{Node } b: &\quad Q^F_{xb},\ Q^F_{yb},\ Q^F_{mb} \\ \text{Node } e: &\quad Q^F_{xe},\ Q^F_{ye},\ Q^F_{me} \end{aligned}$$

Member 1 (MT = 2)

Member 1 has a **hinge at end** e, so it is **MT = 2**.

All stiffness terms are expressed in **kN** and **m**.

Local stiffness matrix (MT = 2):

$$\mathbf{k}_1 = \begin{bmatrix} 260{,}000 & 0 & 0 & -260{,}000 & 0 & 0 \\ 0 & 720 & 3{,}600 & 0 & -720 & 0 \\ 0 & 3{,}600 & 18{,}000 & 0 & -3{,}600 & 0 \\ -260{,}000 & 0 & 0 & 260{,}000 & 0 & 0 \\ 0 & -720 & -3{,}600 & 0 & 720 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Local fixed-end forces for uniform transverse load $w = 19.2\ \mathrm{kN/m}$:

First, fully-fixed (rigid) FEF component.

No inclined force or member, axial = 0:

$$Q_{xb}^F = Q_{xe}^F = 0$$

Shear and moment:

$$Q_{yb}^F = Q_{ye}^F = \frac{wL}{2} = \frac{19.2(5)}{2} = 48\ \mathrm{kN}$$

$$Q_{mb}^F = -Q_{me}^F = \frac{wL^2}{12} = \frac{19.2(5)^2}{12} = 40\ \mathrm{kN\text{-}m}$$

Then modify for MT = 2 (hinge at end e):

$$\mathbf{Q}_1^F = \begin{bmatrix} Q_{xb}^F \\ Q_{yb}^F - \frac{3}{2L} F_{me}^F \\ Q_{mb}^F - \frac{1}{2} F_{me}^F \\ Q_{xe}^F \\ Q_{ye}^F + \frac{3}{2L} F_{me}^F \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 60 \\ 60 \\ 0 \\ 36 \\ 0 \end{bmatrix}$$

Transformation matrix: member 1 is vertical ($\theta = 90^\circ$, $c = 0$, $s = 1$)

$$\mathbf{T}_1 = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Local → Global Map

$$[1, 2, 3, 4, 5, 6]_{\text{local}} = [1, 2, 3, 4, 5, 6]_{\text{global}}$$

Member 2 (MT = 1)

Member 2 has a **hinge at end** b, so it is **MT = 1**.

All stiffness terms are expressed in **kN** and **m**.

Local stiffness matrix (MT = 1):

$$\mathbf{k}_2 = \begin{bmatrix} 260{,}000 & 0 & 0 & -260{,}000 & 0 & 0 \\ 0 & 720 & 0 & 0 & -720 & 3{,}600 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ -260{,}000 & 0 & 0 & 260{,}000 & 0 & 0 \\ 0 & -720 & 0 & 0 & 720 & -3{,}600 \\ 0 & 3{,}600 & 0 & 0 & -3{,}600 & 18{,}000 \end{bmatrix}$$

Local fixed-end forces (point load case shown in the example):

Rigid-member (fully fixed) FEF components used in the example:

$$Q_{xb}^F = Q_{xe}^F = 0$$

$$Q_{yb}^F = Q_{ye}^F = 150\text{kN}$$

$$Q_{mb}^F = -Q_{me}^F = 187.5\text{kN-m}$$

Modify for MT = 1 (hinge at end b):

$$\mathbf{Q}_2^F = \begin{bmatrix} Q_{xb}^F \\ Q_{yb}^F - \frac{3}{2L} Q_{mb}^F \\ 0 \\ Q_{xe}^F \\ Q_{ye}^F + \frac{3}{2L} Q_{mb}^F \\ Q_{me}^F - \frac{1}{2} Q_{mb}^F \end{bmatrix} = \begin{bmatrix} 0 \\ 93.75 \\ 0 \\ 0 \\ 206.25 \\ -281.25 \end{bmatrix}$$

Transformation matrix: member 2 is horizontal (aligned with global), so

$$\mathbf{T}_2 = \mathbf{I}_{6\times 6}$$

$$\mathbf{T}_2 = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Local → Global Map

$$[1, 2, 3, 4, 5, 6]_{\text{local}} = [2, 3, 4, 5, 6, 7]_{\text{global}}$$

Member 3 (MT = 1)

At the start of member 3 there is a **pin support**.

We have two equivalent modeling choices:

Option 1 — Standard Frame Element (MT = 0)

- Use the full 6×6 local stiffness matrix (no releases).
- Impose $u_x = 0$ and $u_y = 0$ in the restrained DOF set.
- Leave the rotational DOF θ free.

In this approach, this location remains numerically stable without any additional treatment.

Option 2 — Member Release at Start (MT = 1)

- Model the pin as a **hinge at the start of the element**.
- This removes moment transfer at that end.
- The element is therefore classified as **MT = 1**.

Because a rotational release eliminates bending stiffness at that end, this creates another situation where fictitous fixity must be applied

In this example, Option 2 is convenient because:

- Geometrically, member 3 matches member 2.
- Same release condition (hinge at b).
- So we can reuse:

$$\mathbf{k}_3 = \mathbf{k}_2$$

- No member loading:

$$\mathbf{Q}_3^F = \mathbf{0}$$

- Vertical member:

$$\theta = 90^\circ \quad \mathbf{T}_3 = \mathbf{T}_2$$

- Mapping

$$[1, 2, 3, 4, 5, 6]_{\text{local}} = [10, 11, 12, 7, 8, 9]_{\text{global}}$$

Note: Both approaches are correct — using MT = 1 simply keeps the setup cleaner.

You will try this process with a standard 6x6 element (in class excercise)

In [1]:
```
import sys
import os
import numpy as np
import pandas as pd

# Get current notebook directory
current_dir = os.getcwd()

# Go up two levels, then into Code/L8
helpers_path = os.path.abspath(
    os.path.join(current_dir, "..", "..", "Code", "L8")
)

sys.path.append(helpers_path)
```

In [2]:
```
from helper_funcs_dsm import (
    assemble_global_stiffness_and_fef,
    partition_system,
    assemble_global_displacements,
    assemble_global_forces
)
from helper_funcs_output import (
    print_element,
    print_dsm_results,
    print_matrix_scaled
)

# -------------------------
# Problem data (hard-coded)
# -------------------------
E = 200e6          # kPa
A = 0.0065         # m^2
I = 150e-6         # m^4
L = 5.0            # m
```

In [3]:

```
# -------------------------
# Member 1 (MT=2)
# -------------------------
k1 = np.array([
    [260000.,     0.,     0., -260000.,     0.,     0.],
    [    0.,   720.,  3600.,      0.,  -720.,     0.],
    [    0.,  3600., 18000.,      0., -3600.,     0.],
    [-260000.,    0.,     0.,  260000.,    0.,     0.],
    [    0.,  -720., -3600.,     0.,   720.,     0.],
    [    0.,     0.,     0.,      0.,     0.,     0.]
], dtype=float)

# Local fixed-end vector for member 1 (given by the example result)
Qf1 = np.array([0., 60., 60., 0., 36., 0.], dtype=float)

# Transformation matrix for theta=90deg -> c=0, s=1
T1 = np.array([
    [ 0.,  1., 0., 0., 0., 0.],
    [-1.,  0., 0., 0., 0., 0.],
    [ 0.,  0., 1., 0., 0., 0.],
    [ 0.,  0., 0., 0., 1., 0.],
    [ 0.,  0., 0.,-1., 0., 0.],
    [ 0.,  0., 0., 0., 0., 1.]
], dtype=float)

# Mapping
m1 = np.array([1,2,3,4,5,6])
```

In [4]:

```
# -------------------------
# Member 2 (MT=1)
# -------------------------
k2 = np.array([
    [260000.,     0.,     0., -260000.,     0.,     0.],
    [    0.,   720.,     0.,      0.,  -720.,  3600.],
    [    0.,     0.,     0.,      0.,     0.,     0.],
    [-260000.,    0.,     0.,  260000.,     0.,     0.],
    [    0.,  -720.,     0.,      0.,   720., -3600.],
    [    0.,  3600.,     0.,      0., -3600., 18000.]
], dtype=float)

# Local fixed-end vector for member 2 (example result)
Qf2 = np.array([0., 93.75, 0., 0., 206.25, -281.25], dtype=float)

# Horizontal member -> no transformation
T2 = np.eye(6, dtype=float)

# Mapping
m2 = np.array([4,5,6,7,8,9])
```

In [5]:

```
# -------------------------
# Member 3 (MT=1)
# -------------------------
k3 = k2.copy()
Qf3 = np.zeros(6, dtype=float)
T3 = T1.copy()
m3 = np.array([10,11,12,7,8,9])
```

In [6]:

```
k_list = [k1, k2, k3]
T_list = [T1, T2, T3]
Qf_list = [Qf1, Qf2, Qf3]
map_list = [m1, m2, m3]   # 1-based DOF maps

# Global system size (still using 1-based maps here)
ndof = int(np.max(np.concatenate(map_list)))

# Initialize Applied Load
F_global = np.zeros(ndof, dtype=float)

# Initialize Global Displacement
u_global = np.zeros(ndof, dtype=float)
```

In [7]:
```
## USER SPECIFIED
F_global[4-1] = 90.75

# Restrained DOFs
dof_restrained_1based = np.array([1, 2, 3, 10, 11], dtype=int)

# Fictitious restrained DOFs
dof_fictitious_1based = np.array([6, 12], dtype=int)

# Combine and sort
dof_restrained_1based = np.sort(
    np.concatenate((dof_restrained_1based, dof_fictitious_1based))
)

print(dof_restrained_1based)
```

```
[ 1  2  3  6 10 11 12]
```

In [8]:
```
K_global, F_fef_global = assemble_global_stiffness_and_fef(
    ndof,
    k_list,
    T_list,
    Qf_list,
    map_list
    )
```

In [9]:
```
print_matrix_scaled(K_global, scale=1000, decimals=1, col_width=3)
```

```
K = 1e+03 ×
1  | 0.7 0.0 -3.6 -0.7 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
2  | 0.0 260.0 0.0 0.0 -260.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
3  | -3.6 0.0 18.0 3.6 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
4  | -0.7 0.0 3.6 260.7 0.0 0.0 -260.0 0.0 0.0 0.0 0.0 0.0
5  | 0.0 -260.0 0.0 0.0 260.7 0.0 0.0 -0.7 3.6 0.0 0.0 0.0
6  | 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
7  | 0.0 0.0 0.0 -260.0 0.0 0.0 260.7 0.0 3.6 -0.7 0.0 0.0
8  | 0.0 0.0 0.0 0.0 -0.7 0.0 0.0 260.7 -3.6 0.0 -260.0 0.0
9  | 0.0 0.0 0.0 0.0 3.6 0.0 3.6 -3.6 36.0 -3.6 0.0 0.0
10 | 0.0 0.0 0.0 0.0 0.0 0.0 -0.7 0.0 -3.6 0.7 0.0 0.0
11 | 0.0 0.0 0.0 0.0 0.0 0.0 0.0 -260.0 0.0 0.0 260.0 0.0
12 | 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0 0.0
```

In [10]:
```
(
    K_ff,
    K_fr,
    K_rf,
    K_rr,
    f_f,
    f_r,
    u_r,
    f_fef_f,
    f_fef_r,
    free_dofs,
    restrained_dofs,
) = partition_system(
    K_global,
    F_global,
    u_global,
    F_fef_global,
    dof_restrained_1based
)
```

$$u_f = K_{ff}^{-1}(F_f - F_f^F - K_{fr}u_r)$$

$$F_r = K_{rf}u_f + K_{rr}u_r + F_r^F$$

In [11]:

```
rhs = f_f - f_fef_f
u_f = np.linalg.solve(K_ff, rhs)

F_r = K_rf @ u_f + f_fef_r

u_global = assemble_global_displacements(
    u_f,
    u_r,
    free_dofs,
    restrained_dofs
    )

f_global_complete = assemble_global_forces(
    f_f,
    F_r,
    free_dofs,
    restrained_dofs
    )
```

In [12]:
```
print_dsm_results(
    u_global,
    f_global_complete,
    dof_restrained_1based,
    dof_fictitious_1based,
    disp_in_mm=True
)
```

```
 DOF  Type     Status  Disp (mm / rad)  Load (kN / kN·m)
   1   u_x      Fixed            0.000          -125.918
   2   u_y      Fixed            0.000            89.168
   3 theta      Fixed            0.000           389.592
   4   u_x       Free           91.553            90.750
   5   u_y       Free           -0.343             0.000
   6 theta Fictitious            0.000             0.000
   7   u_x       Free           91.319             0.000
   8   u_y       Free           -0.811             0.000
   9 theta       Free           -0.001             0.000
  10   u_x      Fixed            0.000           -60.832
  11   u_y      Fixed            0.000           210.832
  12 theta Fictitious            0.000             0.000
```

In [13]:
```
# Element 1,2,3
print_element(1, u_global, m1, T1, k1, Qf1, disp_in_mm=True, dec=1)
print_element(2, u_global, m2, T2, k2, Qf2, disp_in_mm=True, dec=1)
print_element(3, u_global, m3, T3, k3, Qf3, disp_in_mm=True, dec=1)
```

```
E1
u [mm,rad]: [0.0, 0.0, 0.0000, 91.6, -0.3, 0.0000]
v [mm,rad]: [0.0, 0.0, 0.0000, -0.3, -91.6, 0.0000]
q [kN,kN·m]: [89.2, 125.9, 389.6, -89.2, -29.9, 0.0]

E2
u [mm,rad]: [91.6, -0.3, 0.0000, 91.3, -0.8, -0.0014]
v [mm,rad]: [91.6, -0.3, 0.0000, 91.3, -0.8, -0.0014]
q [kN,kN·m]: [60.8, 89.2, 0.0, -60.8, 210.8, -304.2]

E3
u [mm,rad]: [0.0, 0.0, 0.0000, 91.3, -0.8, -0.0014]
v [mm,rad]: [0.0, 0.0, 0.0000, -0.8, -91.3, -0.0014]
q [kN,kN·m]: [210.8, 60.8, 0.0, -210.8, -60.8, 304.2]
```

Part 6 - In-Class Excercise

Open `InClass_Excercise.ipynb` in the `../../Code/L8` folder

Part 7 — Implementing Method B

In [14]:

```
import numpy as np
import pandas as pd

# -------------------------
# Problem data (hard-coded)
# -------------------------
E = 200e6          # kPa
A = 0.0065         # m^2
I = 150e-6         # m^4
L = 5.0            # m
```

In [15]:

```
# -------------------------
# Member 1 (MT=2)
# -------------------------
k1 = np.array([
    [260000.,     0.,     0., -260000.,     0.,     0.],
    [    0.,   720.,  3600.,      0.,  -720.,     0.],
    [    0.,  3600., 18000.,      0., -3600.,     0.],
    [-260000.,    0.,     0.,  260000.,    0.,     0.],
    [    0.,  -720., -3600.,     0.,   720.,     0.],
    [    0.,     0.,     0.,      0.,     0.,     0.]
], dtype=float)

# Local fixed-end vector for member 1 (given by the example result)
Qf1 = np.array([0., 60., 60., 0., 36., 0.], dtype=float)

# Transformation matrix for theta=90deg -> c=0, s=1
T1 = np.array([
    [ 0.,  1., 0., 0., 0., 0.],
    [-1.,  0., 0., 0., 0., 0.],
    [ 0.,  0., 1., 0., 0., 0.],
    [ 0.,  0., 0., 0., 1., 0.],
    [ 0.,  0., 0.,-1., 0., 0.],
    [ 0.,  0., 0., 0., 0., 1.]
], dtype=float)

# Mapping
m1 = np.array([1,2,3,4,5,6])
```

In [16]:

```
# -------------------------
# Member 2 (MT=0)
# -------------------------

factor = (E * I) / (L**3)

k2 = factor * np.array(
    [
        [ (A*L**2)/I,  0.0,      0.0,      -(A*L**2)/I,  0.0,      0.0],
        [ 0.0,         12.0,     6.0*L,    0.0,         -12.0,     6.0*L],
        [ 0.0,         6.0*L,    4.0*L**2, 0.0,         -6.0*L,    2.0*L**2],
        [-(A*L**2)/I,  0.0,      0.0,      (A*L**2)/I,  0.0,      0.0],
        [ 0.0,        -12.0,    -6.0*L,    0.0,          12.0,    -6.0*L],
        [ 0.0,         6.0*L,    2.0*L**2, 0.0,         -6.0*L,    4.0*L**2],
    ],
    dtype=float,
)

# Local fixed-end force vector (MT=0, no releases)
Qf2 = np.array([0.0, 150.0, 187.5, 0.0, 150.0, -187.5], dtype=float)

# Horizontal member -> no transformation
T2 = np.eye(6, dtype=float)

# Mapping
m2 = np.array([4,5,6,7,8,9])
```

In [17]:

```
# -------------------------
# Member 3 (MT=0)
# -------------------------
k3 = k2.copy()

Qf3 = np.zeros(6, dtype=float)
T3 = T1.copy()
m3 = np.array([10,11,12,7,8,9])
```

In [18]:

```
k_list = [k1, k2, k3]
T_list = [T1, T2, T3]
Qf_list = [Qf1, Qf2, Qf3]
map_list = [m1, m2, m3]   # 1-based DOF maps

# Global system size (still using 1-based maps here)
ndof = int(np.max(np.concatenate(map_list)))

# Initialize Applied Load
F_global = np.zeros(ndof, dtype=float)

# Initialize Global Displacement
u_global = np.zeros(ndof, dtype=float)
```

In [19]:

```
## USER SPECIFIED

# Applied Load
F_global[4-1] = 90.75

# Restrained DOFs
dof_restrained_1based = np.array([1, 2, 3, 10, 11], dtype=int)

# Fictitious restrained DOFs
dof_fictitious_1based = np.array([], dtype=int)

# Combine and sort
dof_restrained_1based = np.sort(
    np.concatenate((dof_restrained_1based, dof_fictitious_1based))
)
```

In [20]:
```
K_global, F_fef_global = assemble_global_stiffness_and_fef(
    ndof,
    k_list,
    T_list,
    Qf_list,
    map_list
    )
```

In [21]:
```
(
    K_ff,
    K_fr,
    K_rf,
    K_rr,
    f_f,
    f_r,
    u_r,
    f_fef_f,
    f_fef_r,
    free_dofs,
    restrained_dofs,
) = partition_system(
    K_global,
    F_global,
    u_global,
    F_fef_global,
    dof_restrained_1based
)
```

In [22]:

```
rhs = f_f - f_fef_f
u_f = np.linalg.solve(K_ff, rhs)

F_r = K_rf @ u_f + f_fef_r

u_global = assemble_global_displacements(
    u_f,
    u_r,
    free_dofs,
    restrained_dofs
    )

f_global_complete = assemble_global_forces(
    f_f,
    F_r,
    free_dofs,
    restrained_dofs
    )
```

In [23]:
```
print_dsm_results(
    u_global,
    f_global_complete,
    dof_restrained_1based,
    dof_fictitious_1based,
    disp_in_mm=True
)
```

```
 DOF  Type Status  Disp (mm / rad)  Load (kN / kN·m)
   1   u_x  Fixed            0.000          -125.918
   2   u_y  Fixed            0.000            89.168
   3 theta  Fixed            0.000           389.592
   4   u_x   Free           91.553            90.750
   5   u_y   Free           -0.343             0.000
   6 theta   Free           -0.007             0.000
   7   u_x   Free           91.319             0.000
   8   u_y   Free           -0.811             0.000
   9 theta   Free           -0.001             0.000
  10   u_x  Fixed            0.000           -60.832
  11   u_y  Fixed            0.000           210.832
  12 theta   Free           -0.027             0.000
```

In [24]:
```
# Element 1,2,3
print_element(1, u_global, m1, T1, k1, Qf1, disp_in_mm=True, dec=1)
print_element(2, u_global, m2, T2, k2, Qf2, disp_in_mm=True, dec=1)
print_element(3, u_global, m3, T3, k3, Qf3, disp_in_mm=True, dec=1)
```

```
E1
u [mm,rad]: [0.0, 0.0, 0.0000, 91.6, -0.3, -0.0073]
v [mm,rad]: [0.0, 0.0, 0.0000, -0.3, -91.6, -0.0073]
q [kN,kN·m]: [89.2, 125.9, 389.6, -89.2, -29.9, 0.0]

E2
u [mm,rad]: [91.6, -0.3, -0.0073, 91.3, -0.8, -0.0014]
v [mm,rad]: [91.6, -0.3, -0.0073, 91.3, -0.8, -0.0014]
q [kN,kN·m]: [60.8, 89.2, 0.0, -60.8, 210.8, -304.2]

E3
u [mm,rad]: [0.0, 0.0, -0.0267, 91.3, -0.8, -0.0014]
v [mm,rad]: [0.0, 0.0, -0.0267, -0.8, -91.3, -0.0014]
q [kN,kN·m]: [210.8, 60.8, -0.0, -210.8, -60.8, 304.2]
```

Wrap-Up

In this lecture, we:

- Defined **member releases** and enforced hinge conditions ($M_b = 0$ and/or $M_e = 0$)
- Standardized FEF notation:
  - $F^F_{xb}, F^F_{yb}, F^F_{mb}, F^F_{xe}, F^F_{ye}, F^F_{me}$
- Derived modified member relations for:
  - **MT = 1** (hinge at b)
  - **MT = 2** (hinge at e)
  - **MT = 3** (hinges at both ends)
- Discussed **hinged joints** and why they can make $\mathbf{K}$ singular

Next up:

- Secondary effects (support settlement, temperature, fabrication errors)
- How they enter DSM through equivalent nodal effects / modified $\mathbf{Q}_f$